Assignment - 2

Netflix Data Analysis

Objective:

The objective of this assignment is to analyze Netflix dataset to identify patterns in content type, genres, countries, duration, and user targeting strategies. The goal is to extract meaningful insights that help Netflix decide what type of content to produce and how to expand its business globally.

Methodology:

In this project, Netflix dataset was analyzed using Python. First, the dataset was loaded and explored to understand its structure and key features. Missing values and inconsistent data were handled during the data cleaning stage.

After cleaning, various analysis techniques were applied to answer business-oriented questions such as content distribution, genre popularity, country-wise production, and user targeting strategies. Visualizations like bar charts and line graphs were used to better understand trends and patterns.

This analysis helps in identifying content strategies, audience preferences, and growth opportunities for Netflix.

Tools and Technologies Used:

- Python – Used as the main programming language for data analysis.
- Pandas – Used for data loading, cleaning, and manipulation.
- NumPy – Used for basic numerical operations.
- Matplotlib – Used to create graphs and visualizations.
- Seaborn – Used for improving the visual appearance of graphs.
- Google Colab / Jupyter Notebook – Used as the platform to run the code and perform analysis.

In [31]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style("whitegrid")
```

In [32]:
```
df = pd.read_csv("netflix.csv")
```

In [33]:
```
df.head()
```

Out[33]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	dura
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	Sea
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Sea
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Sea
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	Sea

Dataset Description:

The dataset contains information about movies and TV shows available on Netflix.

- show_id → Unique ID
- type → Movie or TV Show
- title → Name of content
- director → Director name
- cast → Actors
- country → Production country
- date_added → Date added on Netflix
- release_year → Release year
- rating → Content rating
- duration → Duration (minutes/seasons)

- listed_in → Genre
- description → Summary

In [34]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

In [35]:
```
df.isnull().sum()
```

Out[35]:

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

In [36]:
```
df['director'] = df['director'].fillna("Unknown")
df['cast'] = df['cast'].fillna("Unknown")
df['country'] = df['country'].fillna("Unknown")
df['rating'] = df['rating'].fillna("Unknown")
df['date_added'] = pd.to_datetime(df['date_added'], format='mixed')
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month
```

In [37]:
```
df[['date_added','year_added','month_added']].head()
```

Out[37]:

	date_added	year_added	month_added
0	2021-09-25	2021.0	9.0
1	2021-09-24	2021.0	9.0
2	2021-09-24	2021.0	9.0
3	2021-09-24	2021.0	9.0
4	2021-09-24	2021.0	9.0

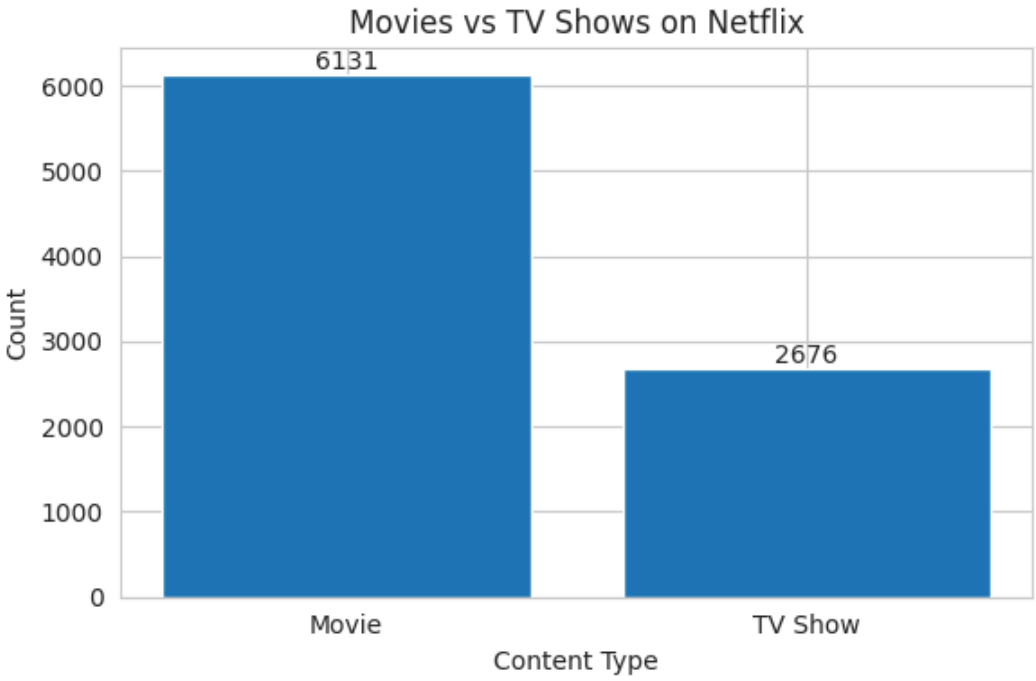
Business Oriented Analysis Questions

1. Content Strategy-

The following business-oriented questions are analyzed using the dataset to derive meaningful insights.

Q1- What is the ratio of Movies vs TV Shows on Netflix?

In [38]:
```
type_counts = df['type'].value_counts()
plt.figure(figsize=(6,4))
plt.bar(type_counts.index, type_counts.values)
plt.title("Movies vs TV Shows on Netflix")
plt.xlabel("Content Type")
plt.ylabel("Count")
for i, v in enumerate(type_counts.values):
    plt.text(i, v, str(v), ha='center', va='bottom')
plt.tight_layout()
plt.show()
ratio = type_counts / type_counts.sum()
print(ratio)
```



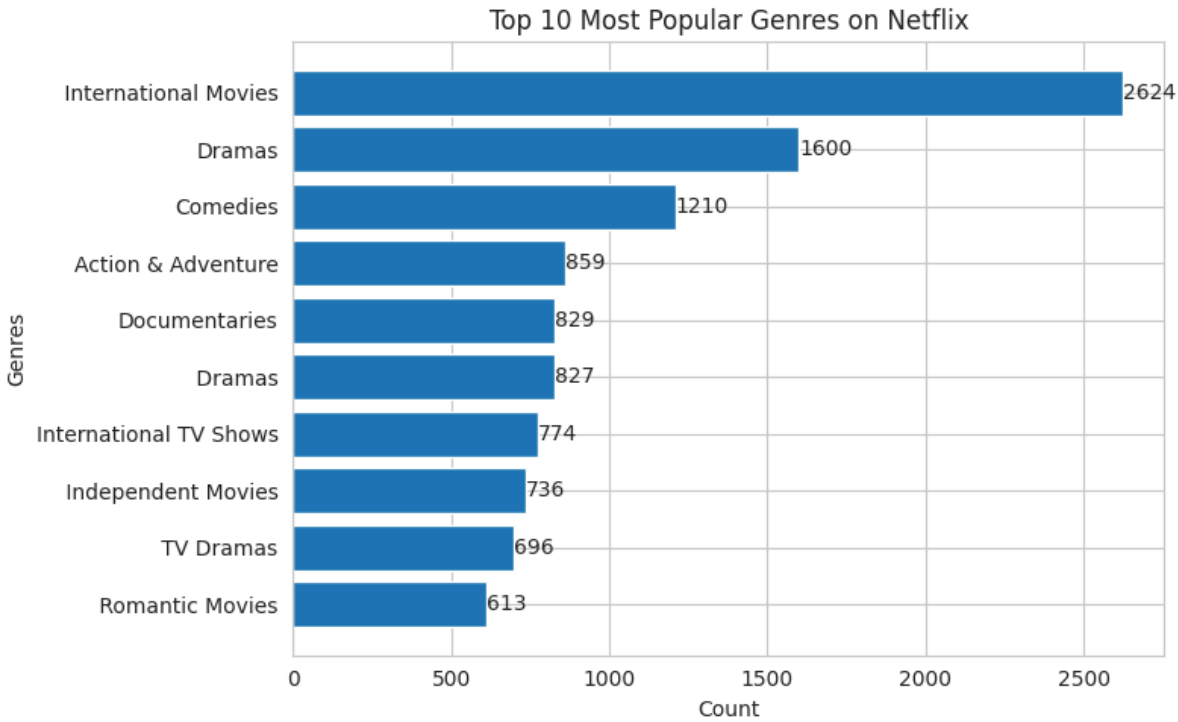

```
type
Movie      0.696151
TV Show    0.303849
Name: count, dtype: float64
```

Insight:

- Movies make up a larger portion of Netflix content compared to TV Shows.
- The ratio shows that Netflix focuses more on movies for quick and diverse content availability.
- TV Shows, although fewer, are important for long-term user engagement and retention.
- The higher number of movies suggests easier production and faster content release cycles.
- Netflix balances both formats to cater to different audience preferences and viewing habits.
- The ratio (~70% movies, ~30% TV shows) clearly shows Netflix's content distribution strategy.

Q2- Which genres are most popular on Netflix globally?

In [39]:
```
genres = df['listed_in'].str.split(',').explode()
genre_counts = genres.value_counts().head(10)
plt.figure(figsize=(8,5))
plt.barh(genre_counts.index, genre_counts.values)
plt.title("Top 10 Most Popular Genres on Netflix")
plt.xlabel("Count")
plt.ylabel("Genres")
plt.gca().invert_yaxis()
for i, v in enumerate(genre_counts.values):
    plt.text(v, i, str(v), va='center')
plt.tight_layout()
plt.show()
```



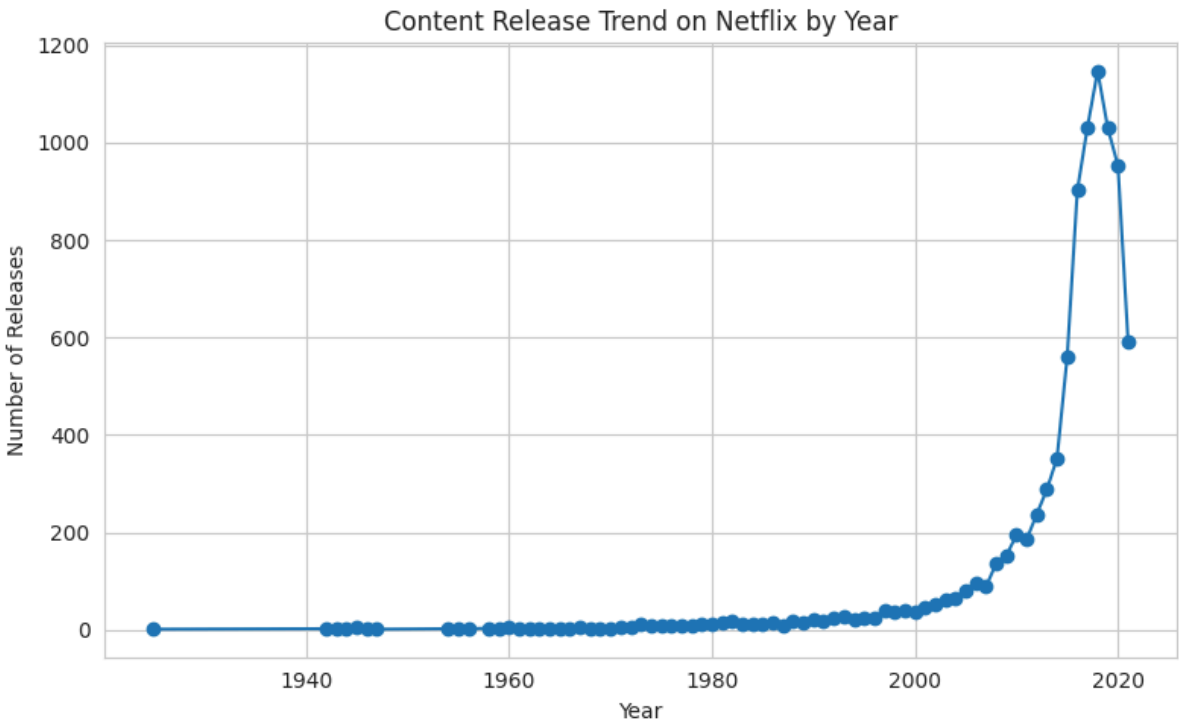
Insight:

- Drama is the most popular genre on Netflix, indicating its strong appeal across global audiences.
- Comedy and International content are also among the top genres, showing demand for entertainment and cultural diversity.
- The presence of multi-genre content highlights Netflix's strategy of combining genres to attract wider audiences.

- Popular genres reflect viewer preferences and should be prioritized for future content production.
- The dominance of these genres suggests that emotional storytelling and relatable content drive higher engagement.

Q3- Which years saw the highest release of content on Netflix?

In [40]:
```
year_counts = df['release_year'].value_counts().sort_index()
plt.figure(figsize=(8,5))
plt.plot(year_counts.index, year_counts.values, marker='o')
plt.title("Content Release Trend on Netflix by Year")
plt.xlabel("Year")
plt.ylabel("Number of Releases")
plt.tight_layout()
plt.show()
```



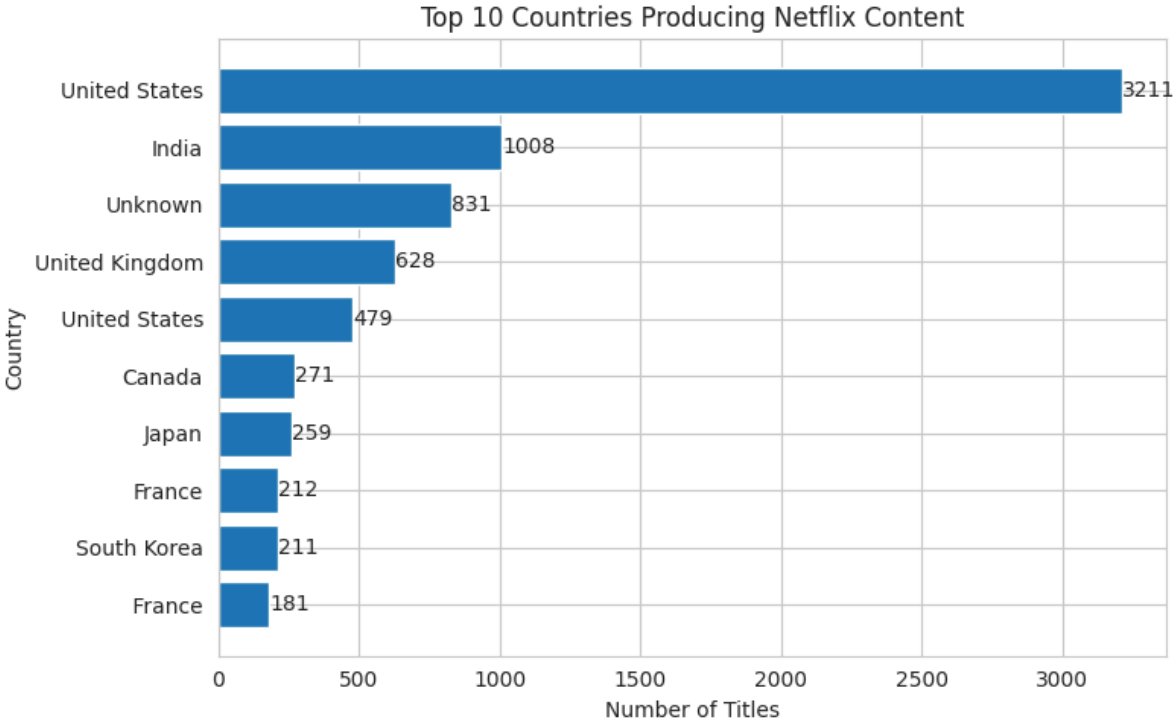
Insight:

- The number of content releases increased significantly after 2015.
- The highest releases are observed in recent years, indicating rapid expansion of Netflix.
- This growth reflects Netflix's aggressive investment in original and licensed content.
- Increasing releases align with the rise in global subscribers and competition in the streaming industry.
- The trend shows that Netflix continuously adds fresh content to keep users engaged and reduce churn.

Q4- Which countries produce the most Netflix content?

In [41]:
```
countries = df['country'].str.split(',').explode()
country_counts = countries.value_counts().head(10)
plt.figure(figsize=(8,5))
plt.barh(country_counts.index, country_counts.values)
```

```
plt.title("Top 10 Countries Producing Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Country")
plt.gca().invert_yaxis()
for i, v in enumerate(country_counts.values):
    plt.text(v, i, str(v), va='center')
plt.tight_layout()
plt.show()
```

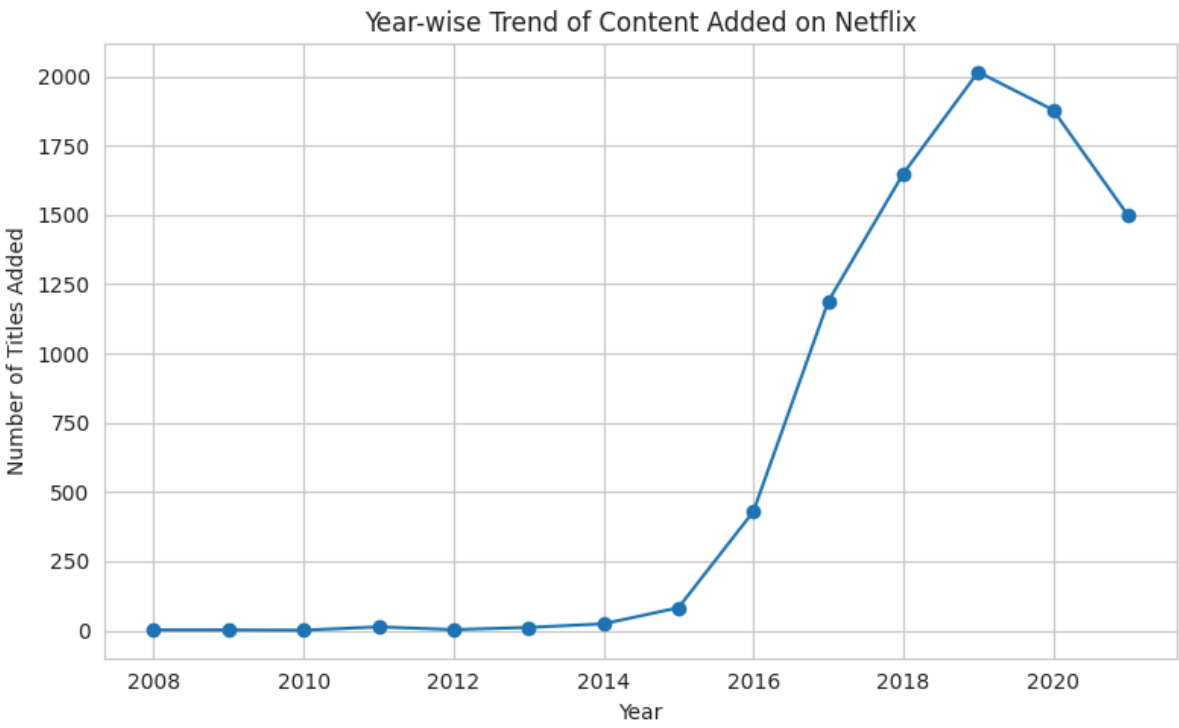


Insight:

- The United States produces the highest amount of content on Netflix, making it the dominant market.
- India and the United Kingdom are also among the top content-producing countries.
- High production from these countries reflects strong film and television industries.
- Netflix relies on both global (USA) and regional (India, UK) markets for content diversity.
- Country-wise distribution shows Netflix's strategy of expanding into international markets to attract a wider audience.

Q5- How has the trend of adding new content evolved year by year?

In [42]:
```
df['year_added'] = df['date_added'].dt.year
year_added_counts = df['year_added'].value_counts().sort_index()
plt.figure(figsize=(8,5))
plt.plot(year_added_counts.index, year_added_counts.values, marker='o')
plt.title("Year-wise Trend of Content Added on Netflix")
plt.xlabel("Year")
plt.ylabel("Number of Titles Added")
plt.tight_layout()
plt.show()
```



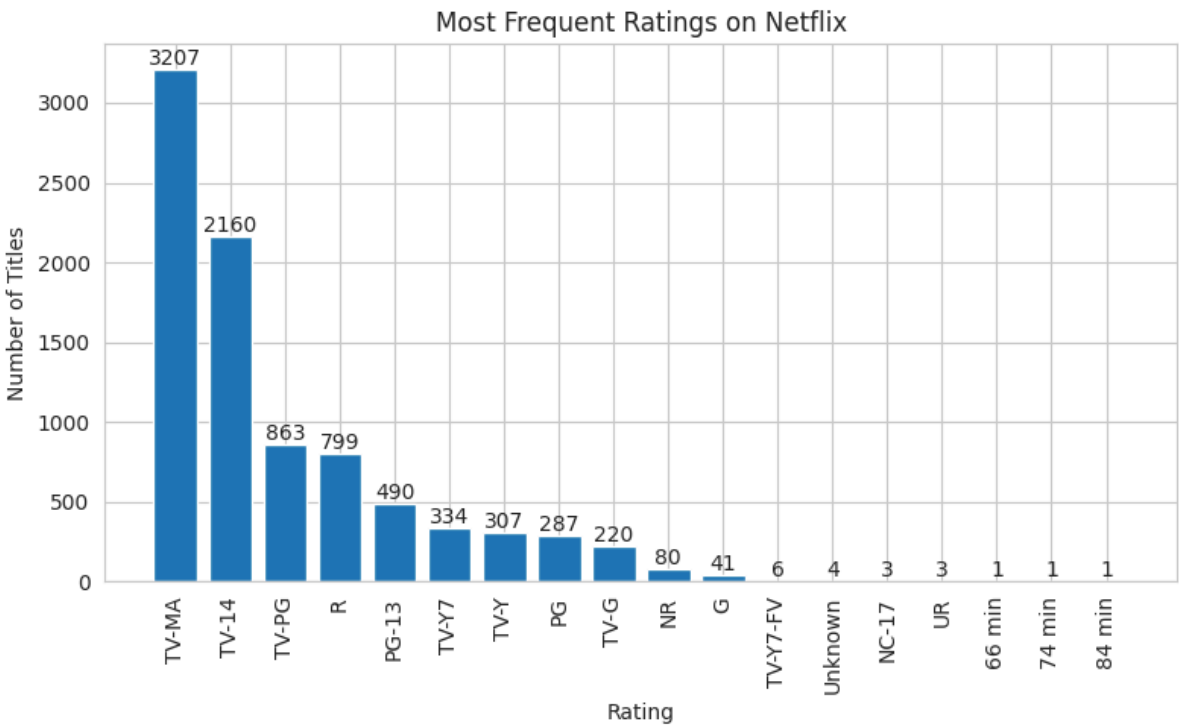
Insight:

- The number of titles added to Netflix increased rapidly after 2015.
- A sharp growth trend indicates Netflix's aggressive global expansion strategy.
- The platform focused heavily on adding new content to attract and retain subscribers.
- Recent years show peak content addition, reflecting high competition in the streaming industry.
- Continuous addition of content suggests Netflix's strategy to keep the platform fresh and engaging for users.

2. User Demographics & Targeting-

Q6- Which ratings are most frequent on Netflix?

In [43]:
```
rating_counts = df['rating'].value_counts()
plt.figure(figsize=(8,5))
plt.bar(rating_counts.index, rating_counts.values)
plt.title("Most Frequent Ratings on Netflix")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=90)
for i, v in enumerate(rating_counts.values):
    plt.text(i, v, str(v), ha='center', va='bottom')
plt.tight_layout()
plt.show()
```

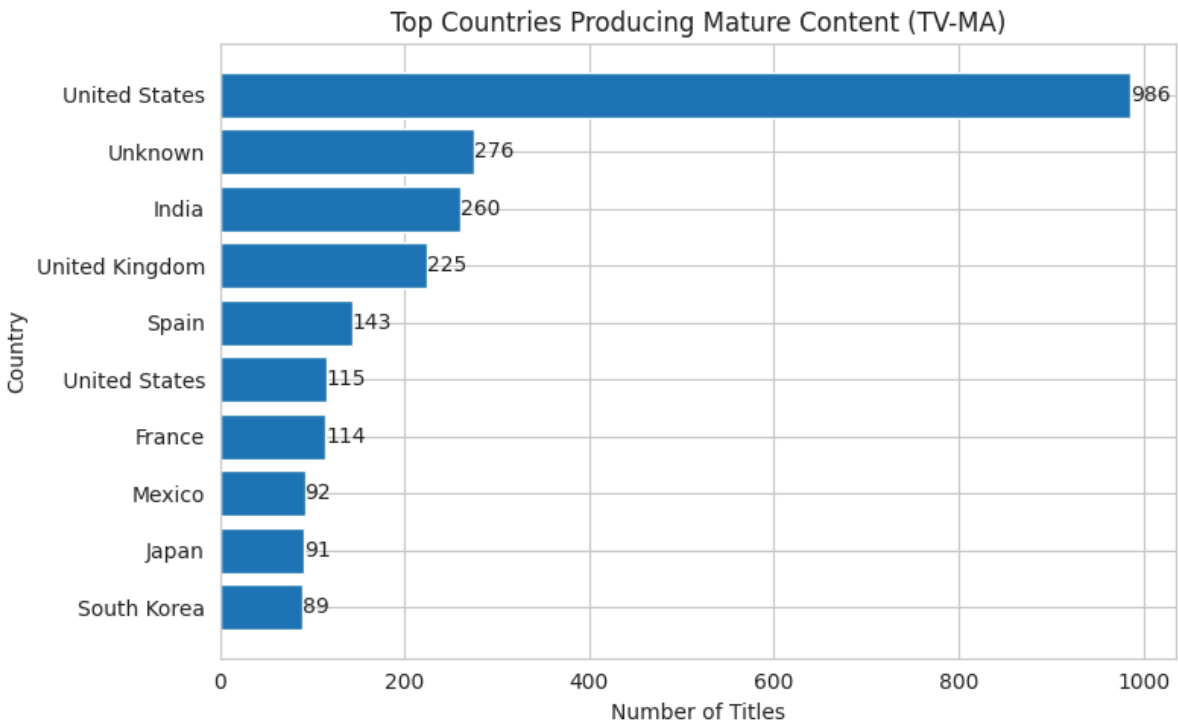


Insight:

- TV-MA is the most frequent rating on Netflix, indicating a strong focus on mature audience content.
- Ratings like TV-14 and PG-13 are also common, showing that Netflix caters to teenagers and young adults.
- The presence of multiple rating categories highlights Netflix's effort to target diverse age groups.
- A higher proportion of mature content suggests growing demand for realistic and bold storytelling.
- The distribution of ratings reflects Netflix's balanced strategy between adult-focused and family-friendly content.

Q7- Do some countries tend to produce more mature content (TV-MA)?

In [44]:
```
mature_content = df[df['rating'] == 'TV-MA']
mature_countries = mature_content['country'].str.split(',').explode()
mature_counts = mature_countries.value_counts().head(10)
plt.figure(figsize=(8,5))
plt.barh(mature_counts.index, mature_counts.values)
plt.title("Top Countries Producing Mature Content (TV-MA)")
plt.xlabel("Number of Titles")
plt.ylabel("Country")
plt.gca().invert_yaxis()
for i, v in enumerate(mature_counts.values):
    plt.text(v, i, str(v), va='center')
plt.tight_layout()
plt.show()
```

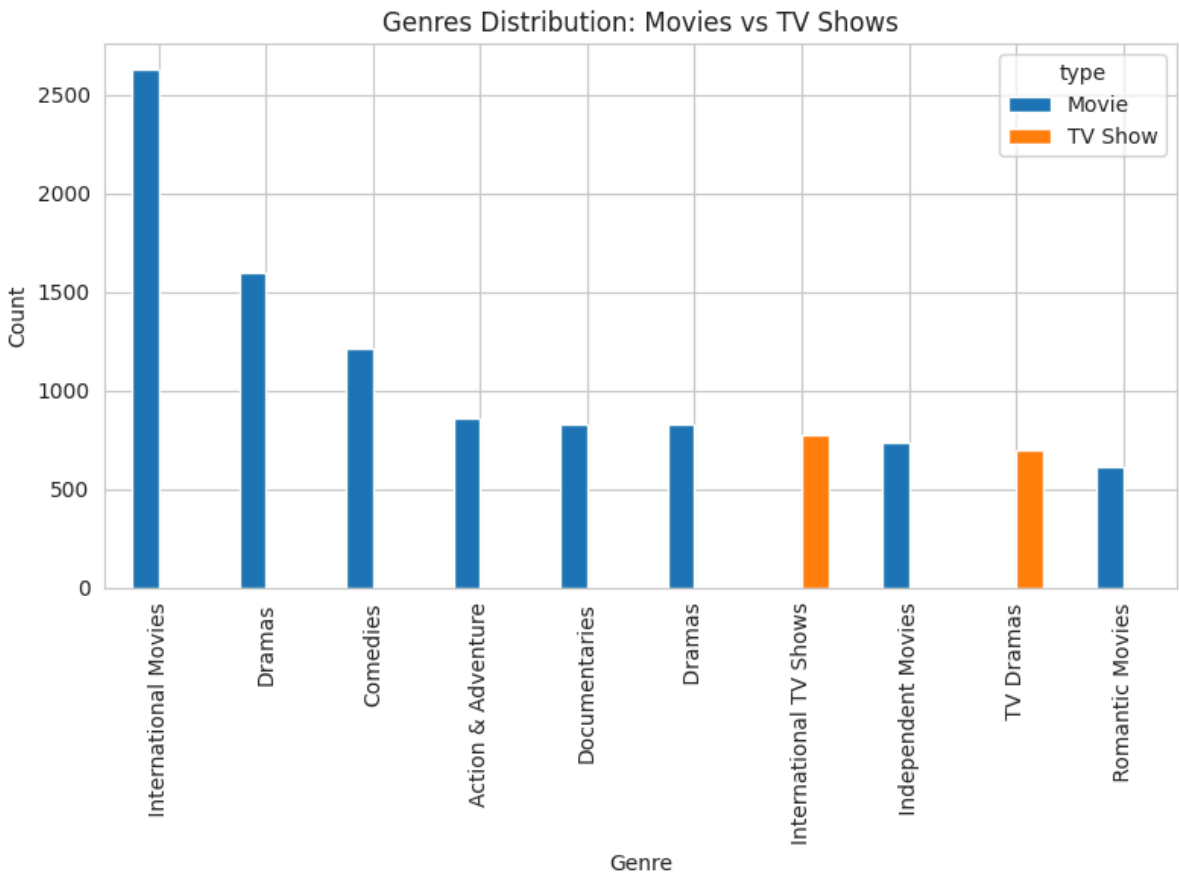


Insight:

- The United States produces the highest amount of mature (TV-MA) content on Netflix.
- Other developed countries like the United Kingdom and Canada also contribute significantly.
- Mature content is more common in regions with higher acceptance of bold and realistic storytelling.
- Cultural and regulatory differences influence the type of content produced in each country.
- The dominance of TV-MA content suggests strong demand for adult-oriented entertainment globally.

Q8- Which genres are more associated with TV Shows vs Movies?

In [45]:
```
df['genre'] = df['listed_in'].str.split(',')
df_exploded = df.explode('genre')
genre_type = df_exploded.groupby(['type', 'genre']).size().unstack().fillna(0)
top_genres = genre_type.sum().sort_values(ascending=False).head(10).index
genre_type[top_genres].T.plot(kind='bar', figsize=(8,6))
plt.title("Genres Distribution: Movies vs TV Shows")
plt.xlabel("Genre")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



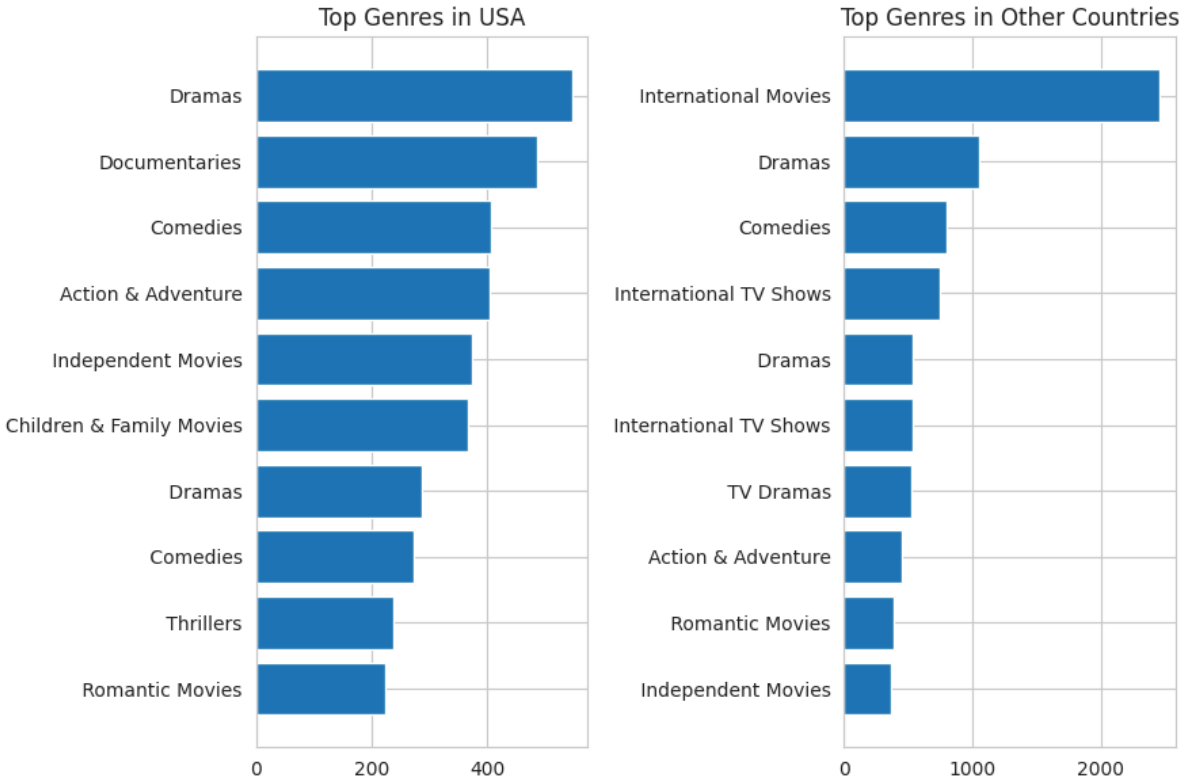
Insight:

- Movies dominate across most genres, especially in categories like Drama, Comedy, and Action.
- TV Shows are more concentrated in genres such as Drama, International TV Shows, and Crime series.
- TV Shows focus more on long-form storytelling, allowing deeper character and plot development.
- Movies provide quick entertainment, making them more diverse across multiple genres.
- The difference in genre distribution highlights Netflix's strategy of balancing short-form and long-form content.

Q9- Which genres dominate the U.S. vs other countries?

In [46]:
```
us_content = df[df['country'].str.contains("United States", na=False)]
non_us_content = df[~df['country'].str.contains("United States", na=False)]
us_genres = us_content['listed_in'].str.split(',').explode()
non_us_genres = non_us_content['listed_in'].str.split(',').explode()
us_top = us_genres.value_counts().head(10)
non_us_top = non_us_genres.value_counts().head(10)
plt.figure(figsize=(9,6))
plt.subplot(1,2,1)
plt.barh(us_top.index, us_top.values)
plt.title("Top Genres in USA")
plt.gca().invert_yaxis()
plt.subplot(1,2,2)
plt.barh(non_us_top.index, non_us_top.values)
plt.title("Top Genres in Other Countries")
plt.gca().invert_yaxis()
```

```
plt.tight_layout()
plt.show()
```

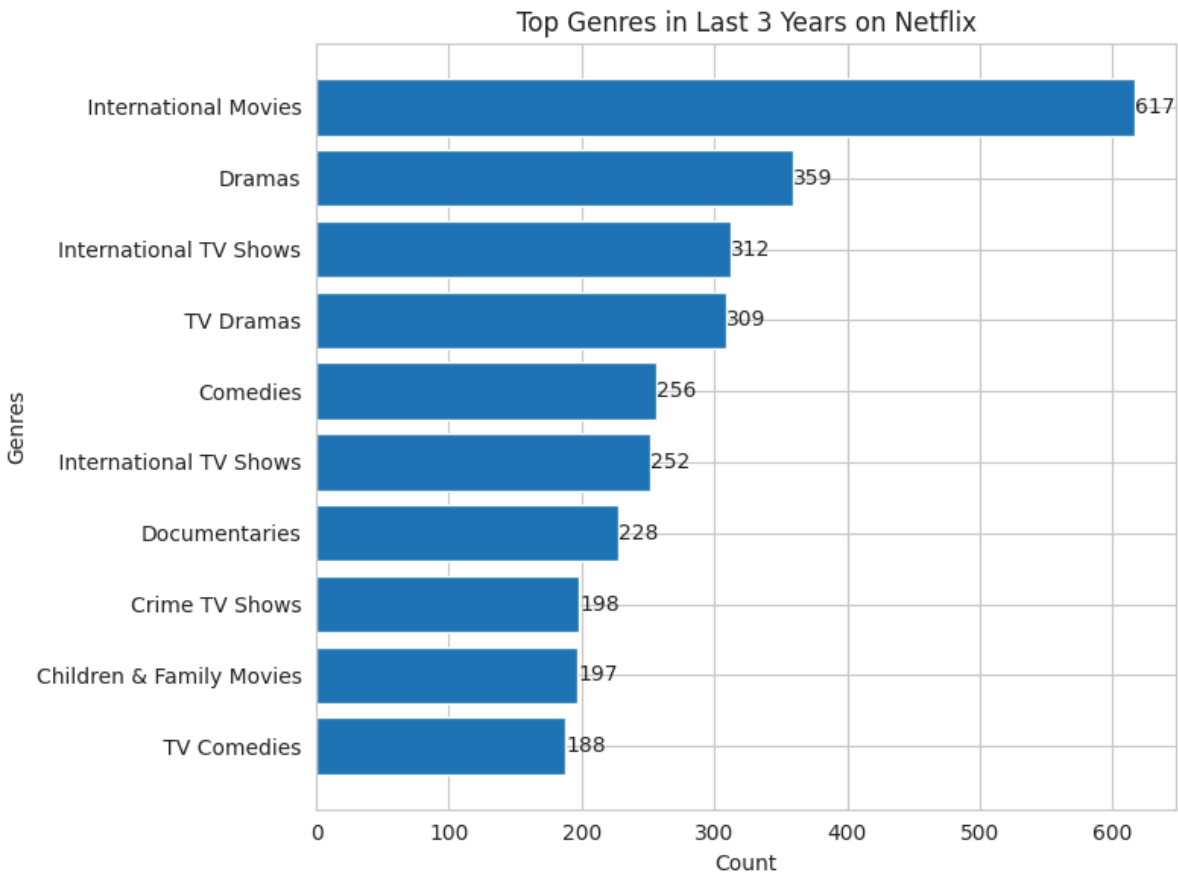


Insight:

- The United States shows a wide variety of genres, with strong dominance in Drama, Comedy, and Action.
- Non-US countries focus more on International Movies, Drama, and regional content.
- US content is more diverse, reflecting a mature and large entertainment industry.
- Other countries emphasize culturally relevant and localized storytelling.
- This difference highlights Netflix's strategy of balancing global content with region-specific preferences.

Q10- What genres are most popular in the last 3 years?

In [47]:
```
recent_year = df['release_year'].max()
recent_data = df[df['release_year'] >= recent_year - 2]
recent_genres = recent_data['listed_in'].str.split(',').explode()
top_recent_genres = recent_genres.value_counts().head(10)
plt.figure(figsize=(8,6))
plt.barh(top_recent_genres.index, top_recent_genres.values)
plt.title("Top Genres in Last 3 Years on Netflix")
plt.xlabel("Count")
plt.ylabel("Genres")
plt.gca().invert_yaxis()
for i, v in enumerate(top_recent_genres.values):
    plt.text(v, i, str(v), va='center')
plt.tight_layout()
plt.show()
```



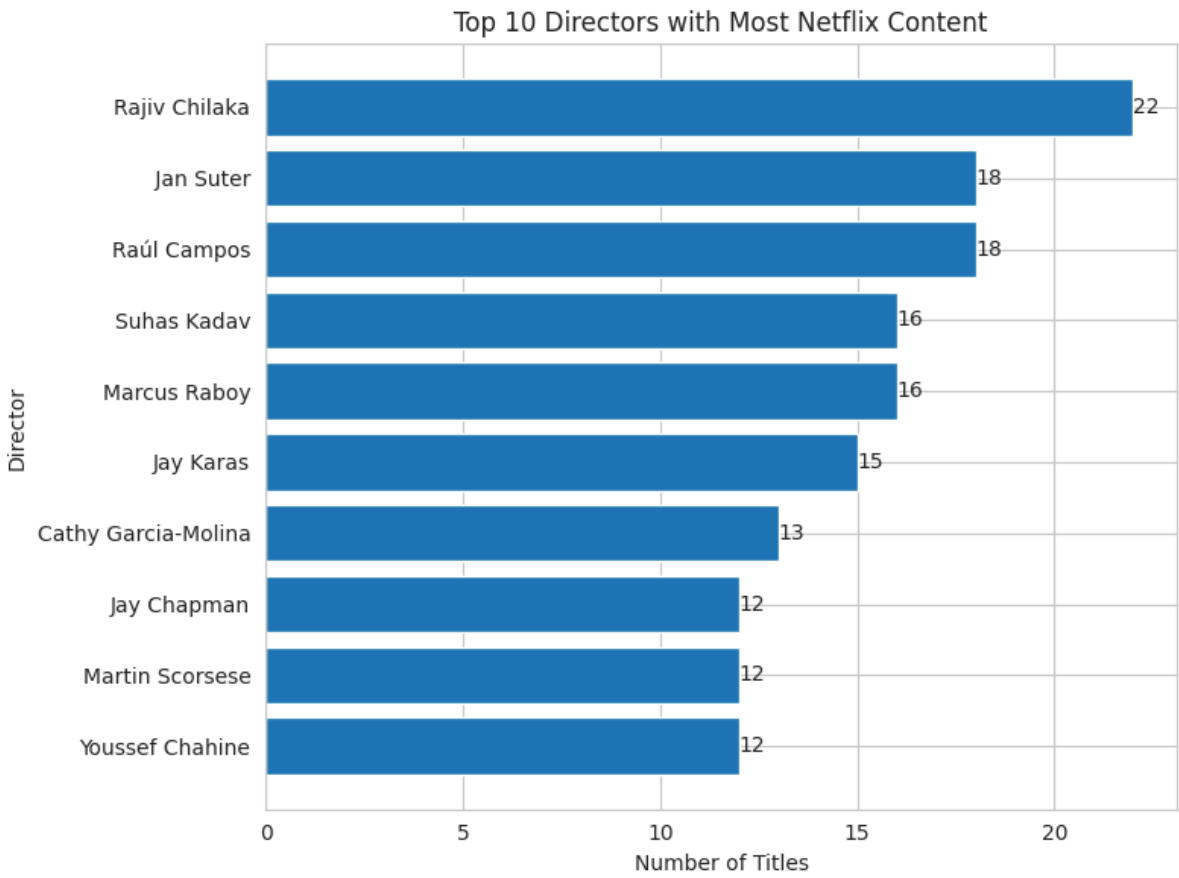
Insight:

- Drama continues to be the most popular genre even in recent years, showing its consistent demand.
- Thriller and Crime genres have gained significant popularity, indicating a shift toward intense and engaging storytelling.
- Documentaries have seen a rise, reflecting audience interest in real-world and informative content.
- International content is growing rapidly, showing Netflix's focus on global expansion.
- Recent trends suggest that viewers prefer realistic, suspenseful, and culturally diverse content.

3. Talent Acquisition & Partnerships-

Q11- Who are the top 10 directors with the most Netflix content?

In [48]:
```
directors = df['director'].str.split(',').explode()
directors = directors[directors != "Unknown"]
top_directors = directors.value_counts().head(10)
plt.figure(figsize=(8,6))
plt.barh(top_directors.index, top_directors.values)
plt.title("Top 10 Directors with Most Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Director")
plt.gca().invert_yaxis()
for i, v in enumerate(top_directors.values):
    plt.text(v, i, str(v), va='center')
plt.tight_layout()
plt.show()
```

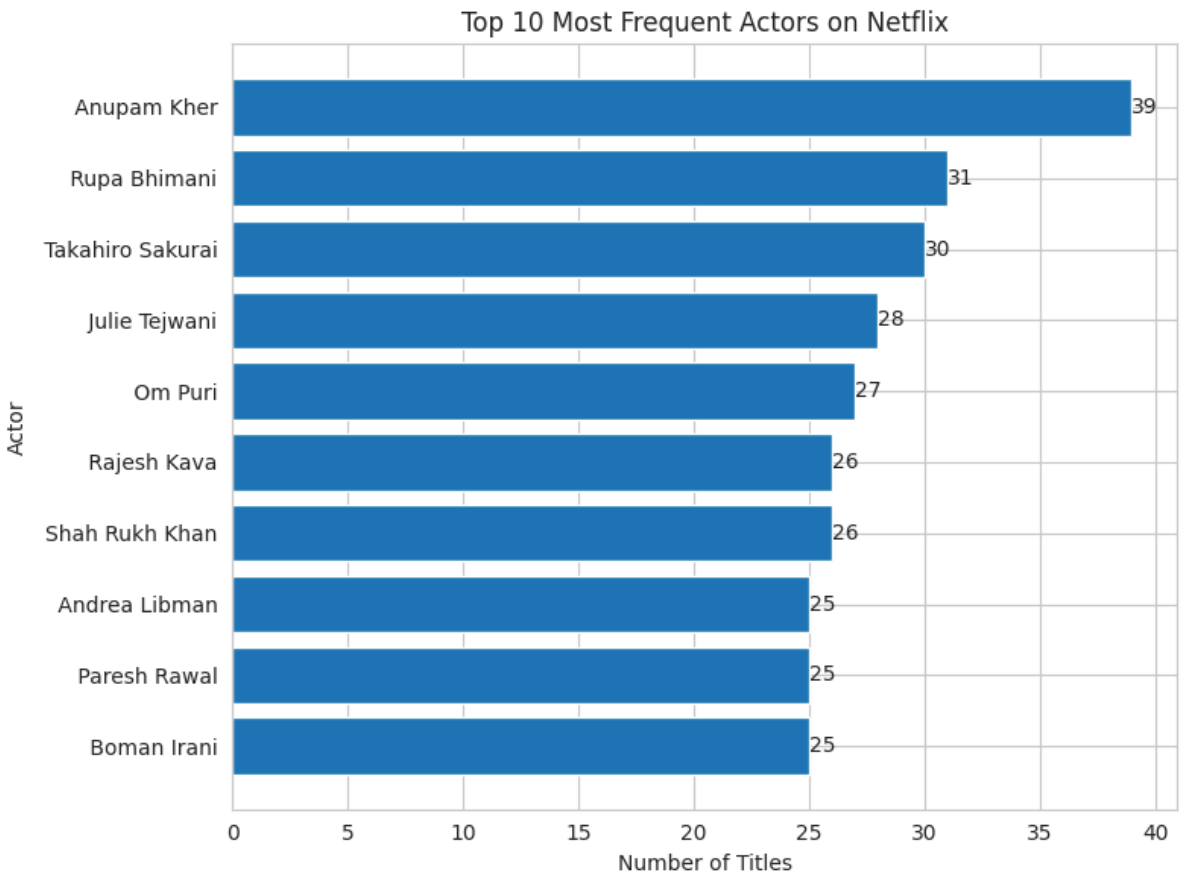


Insight:

- A few directors contribute a large number of titles on Netflix, indicating strong partnerships.
- These top directors are likely trusted creators who consistently deliver successful content.
- Frequent collaborations help Netflix maintain content quality and audience engagement.
- Directors with multiple titles show specialization and reliability in content production.
- Identifying such directors helps Netflix strengthen long-term creative partnerships.

Q12- Which actors appear most frequently in Netflix shows?

In [49]:
```
actors = df['cast'].str.split(',').explode()
actors = actors[actors != "Unknown"]
top_actors = actors.value_counts().head(10)
plt.figure(figsize=(8,6))
plt.barh(top_actors.index, top_actors.values)
plt.title("Top 10 Most Frequent Actors on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Actor")
plt.gca().invert_yaxis()
for i, v in enumerate(top_actors.values):
    plt.text(v, i, str(v), va='center')
plt.tight_layout()
plt.show()
```

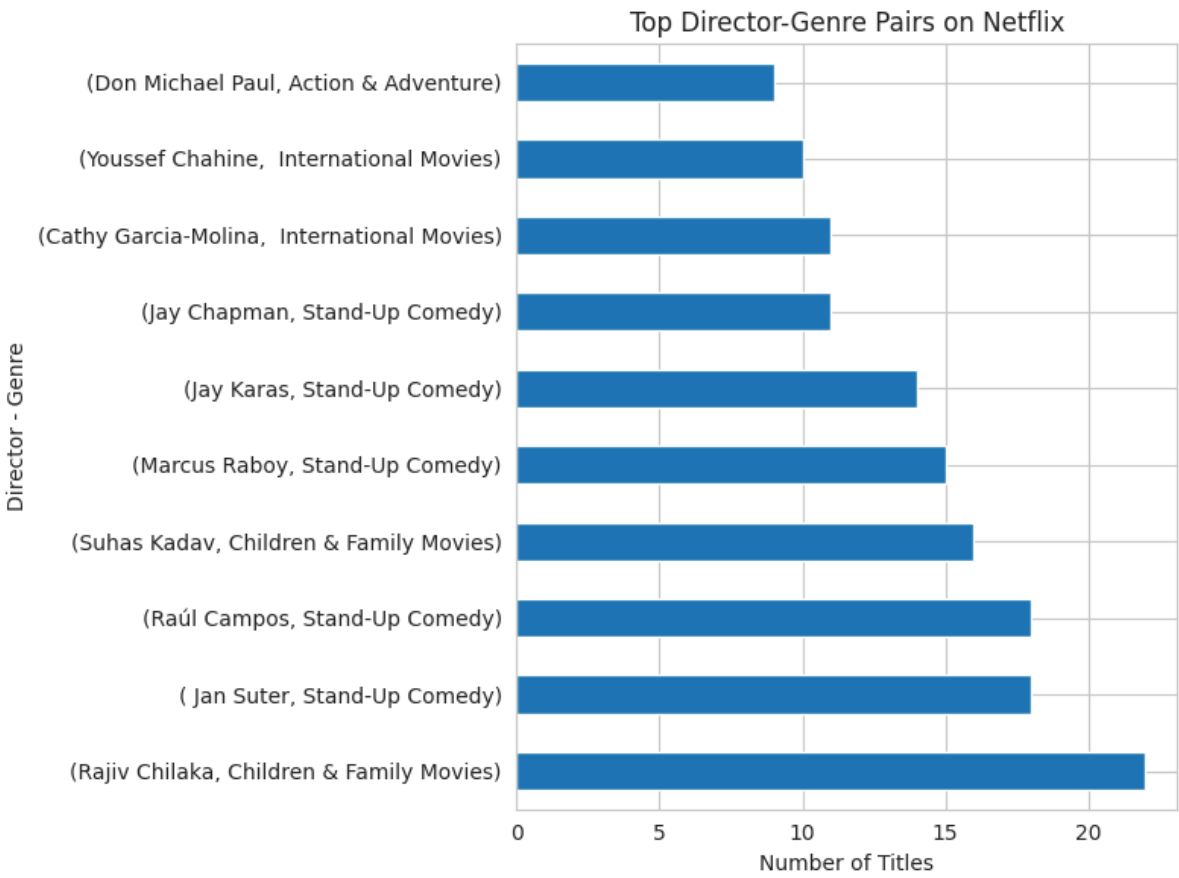


Insight:

- A few actors appear frequently across multiple Netflix titles, indicating strong platform presence.
- Popular actors are likely to attract more viewers and improve content performance.
- Repeated casting suggests ongoing collaborations between Netflix and certain actors.
- High-frequency actors may represent reliable performers in different genres.
- Star power plays a significant role in audience engagement and content popularity.

Q13- Which director-genre pairs are most frequent?

In [50]:
```
df['director_split'] = df['director'].str.split(',')
df['genre_split'] = df['listed_in'].str.split(',')
df_exploded = df.explode('director_split').explode('genre_split')
df_exploded = df_exploded[df_exploded['director_split'] != "Unknown"]
director_genre = df_exploded.groupby(['director_split', 'genre_split']).size().sort_values(
plt.figure(figsize=(8,6))
director_genre.plot(kind='barh')
plt.title("Top Director-Genre Pairs on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Director - Genre")
plt.tight_layout()
plt.show()
```



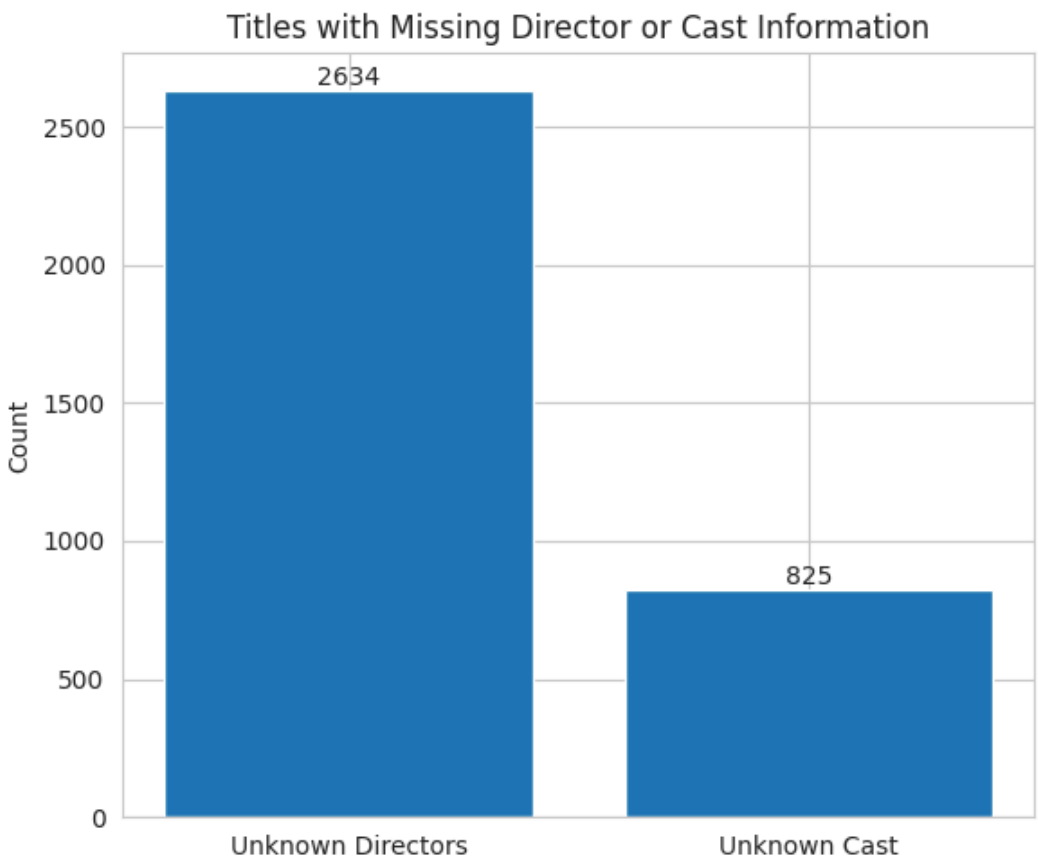
Insight:

- Certain directors consistently work within specific genres, indicating specialization.
- Frequent director-genre pairs show strong expertise and audience preference alignment.
- Directors focusing on popular genres like Drama and Comedy tend to have higher content counts.
- Repeated collaboration in the same genre helps maintain content quality and consistency.
- These patterns can help Netflix identify reliable creators for specific types of content.

Q14- How many titles have unknown directors or cast members?

In [51]:
```
unknown_directors = (df['director'] == "Unknown").sum()
unknown_cast = (df['cast'] == "Unknown").sum()
print("Unknown Directors:", unknown_directors)
print("Unknown Cast:", unknown_cast)
labels = ['Unknown Directors', 'Unknown Cast']
values = [unknown_directors, unknown_cast]
plt.figure(figsize=(6,5))
plt.bar(labels, values)
plt.title("Titles with Missing Director or Cast Information")
plt.ylabel("Count")
for i, v in enumerate(values):
    plt.text(i, v, str(v), ha='center', va='bottom')
plt.tight_layout()
plt.show()
```

```
Unknown Directors: 2634
Unknown Cast: 825
```



Insight:

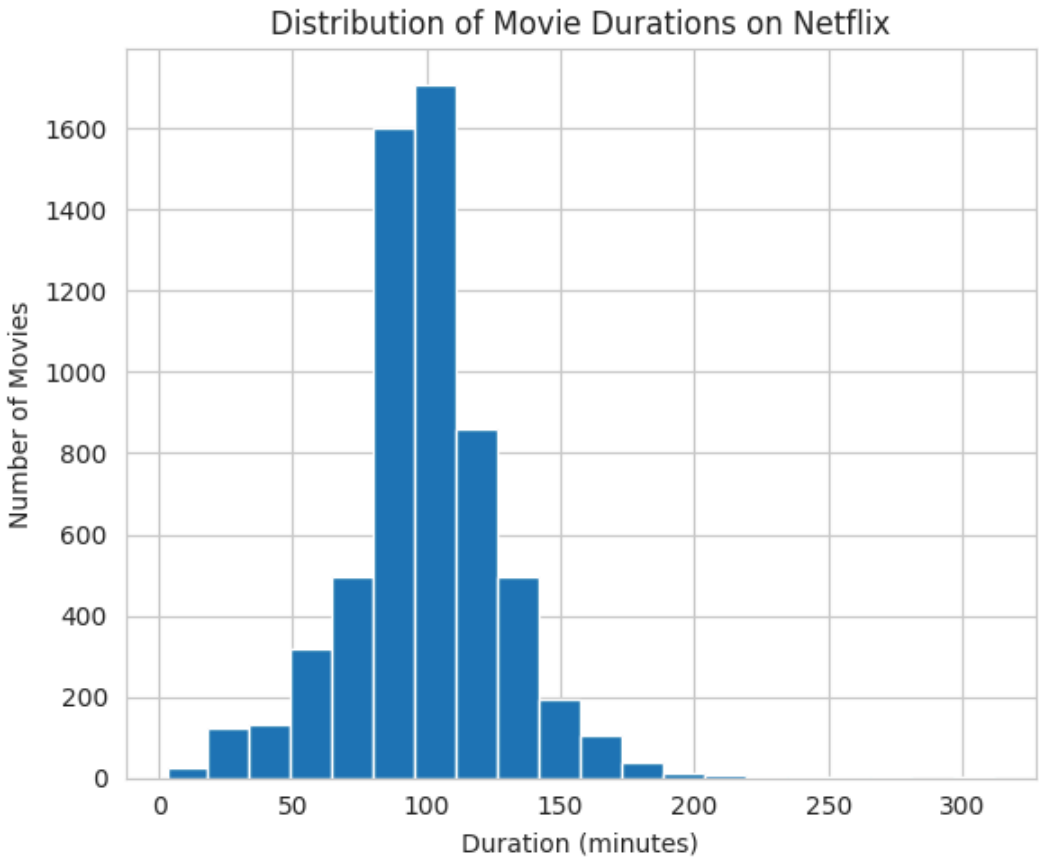
- A noticeable number of titles have missing director information compared to cast details.
- Missing cast data is relatively lower, indicating better availability of actor-related information.
- These data gaps can impact the accuracy of analysis, especially in talent-related insights.
- Incomplete metadata suggests inconsistencies in data collection or content documentation.
- Improving data quality can help Netflix make more accurate content and partnership decisions.

4. Duration & Engagement-

Q15- What is the average duration of Movies on Netflix?

In [52]:
```
movies = df[df['type'] == 'Movie'].copy()
movies['duration'] = movies['duration'].str.replace(" min", "")
movies['duration'] = movies['duration'].astype(float)
avg_duration = movies['duration'].mean()
print("Average Movie Duration:", avg_duration)
plt.figure(figsize=(6,5))
plt.hist(movies['duration'], bins=20)
plt.title("Distribution of Movie Durations on Netflix")
plt.xlabel("Duration (minutes)")
plt.ylabel("Number of Movies")
plt.tight_layout()
plt.show()
```

```
Average Movie Duration: 99.57718668407311
```

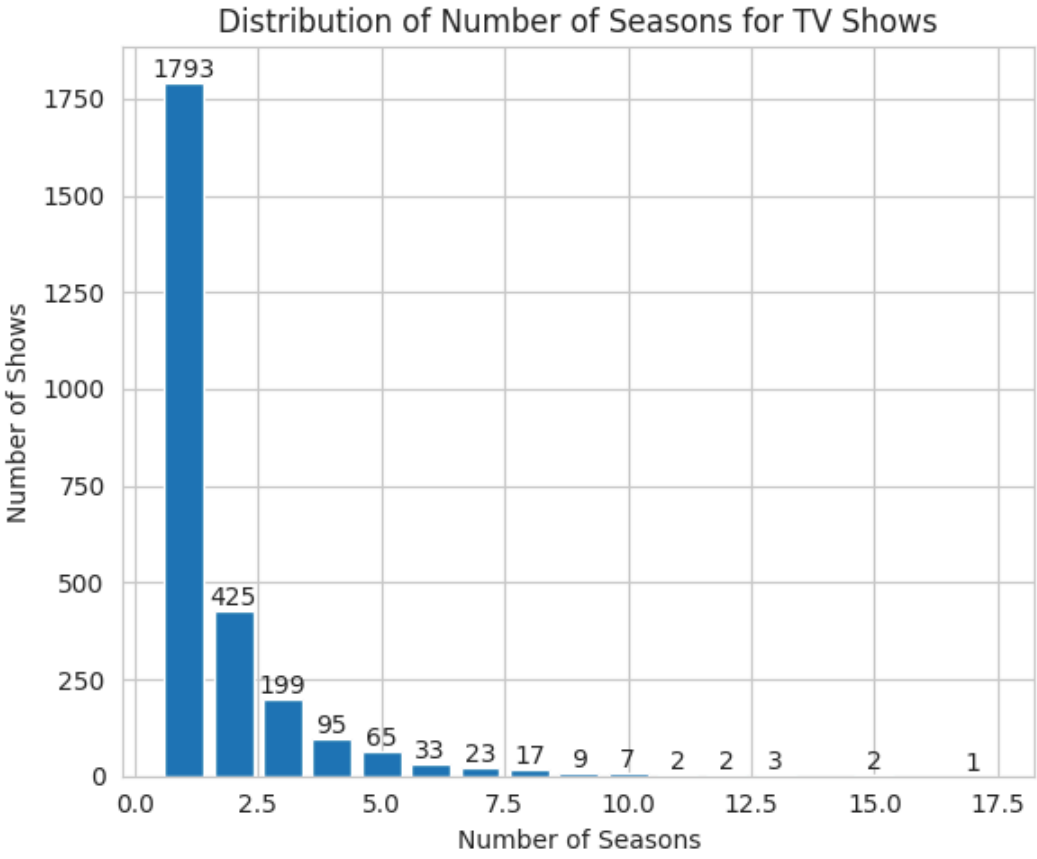


Insight:

- The average movie duration on Netflix is approximately between 90 to 120 minutes.
- Most movies fall within a standard duration range, indicating a consistent content format.
- Extremely short or very long movies are less common on the platform.
- This duration aligns with typical audience attention span and viewing preferences.
- Maintaining optimal movie length helps improve user engagement and completion rates.

Q16- What's the most common number of seasons for TV shows?

In [53]:
```
tv_shows = df[df['type'] == 'TV Show'].copy()
tv_shows['seasons'] = tv_shows['duration'].str.replace(" Seasons", "")
tv_shows['seasons'] = tv_shows['seasons'].str.replace(" Season", "")
tv_shows['seasons'] = tv_shows['seasons'].astype(int)
season_counts = tv_shows['seasons'].value_counts().sort_index()
plt.figure(figsize=(6,5))
plt.bar(season_counts.index, season_counts.values)
plt.title("Distribution of Number of Seasons for TV Shows")
plt.xlabel("Number of Seasons")
plt.ylabel("Number of Shows")
for i, v in enumerate(season_counts.values):
    plt.text(season_counts.index[i], v, str(v), ha='center', va='bottom')
plt.tight_layout()
plt.show()
print("Most Common Seasons:", season_counts.idxmax())
```



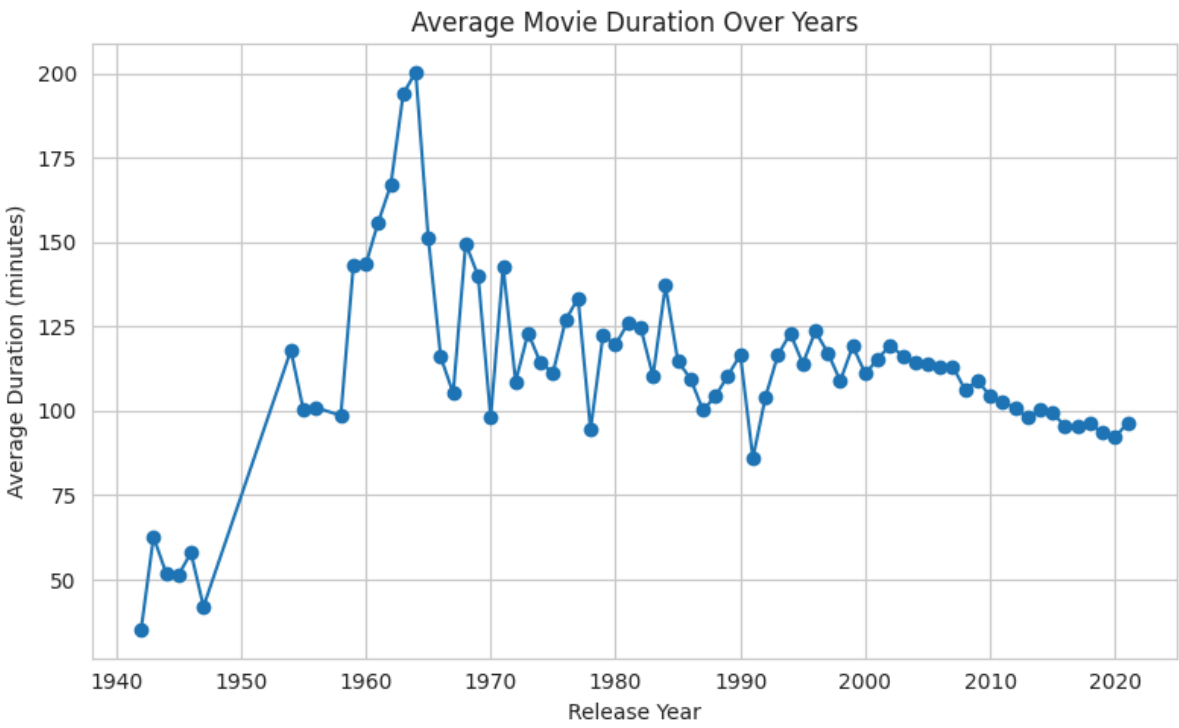

```
Most Common Seasons: 1
```

Insight:

- The most common number of seasons for TV Shows on Netflix is 1 season.
- This indicates a strong trend toward limited series or short-format storytelling.
- Shorter series reduce production cost and allow faster content release cycles.
- Limited seasons help maintain viewer interest without long-term commitment.
- This strategy reflects changing audience preferences toward quick and engaging content.

Q17- Is there a trend in movie durations over the years?

In [54]:
```
movies = df[df['type'] == 'Movie'].copy()
movies['duration'] = movies['duration'].str.replace(" min", "")
movies['duration'] = movies['duration'].astype(float)
duration_trend = movies.groupby('release_year')['duration'].mean()
plt.figure(figsize=(8,5))
plt.plot(duration_trend.index, duration_trend.values, marker='o')
plt.title("Average Movie Duration Over Years")
plt.xlabel("Release Year")
plt.ylabel("Average Duration (minutes)")
plt.tight_layout()
plt.show()
```

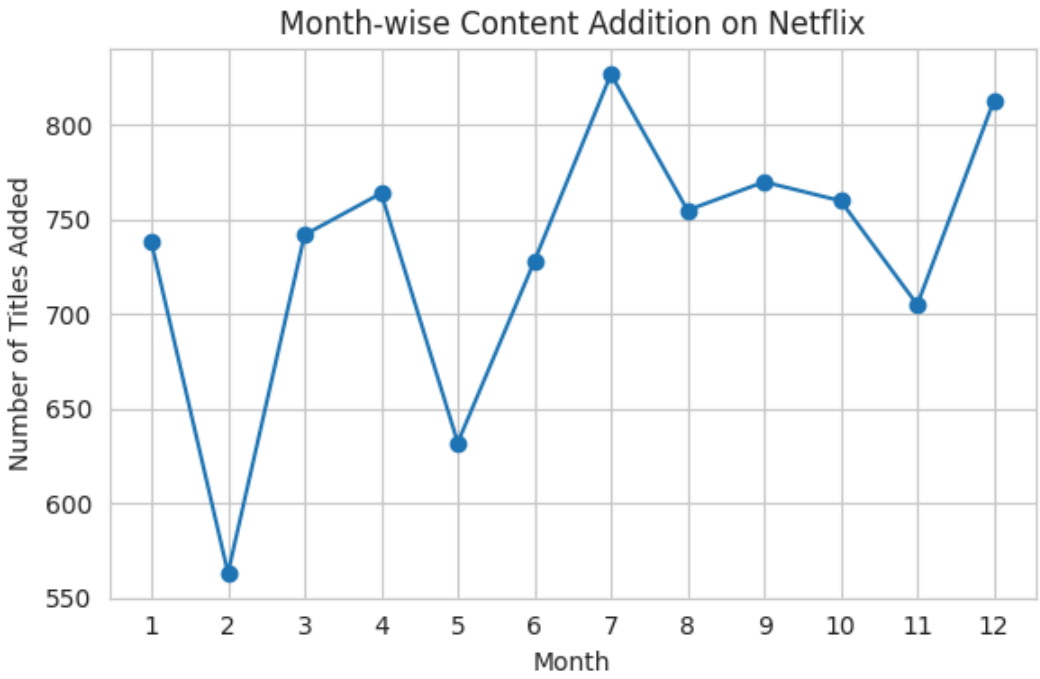


Insight:

- Movie durations have remained relatively stable over the years.
- There is no significant upward or downward trend in average movie length.
- Most movies consistently fall within the standard 90–120 minutes range.
- Stability in duration suggests consistent audience preferences over time.
- Netflix maintains similar content formats to ensure viewer engagement and satisfaction.

Q18- In which months does Netflix add the most content?

In [55]:
```
df['month_added'] = df['date_added'].dt.month
month_counts = df['month_added'].value_counts().sort_index()
plt.figure(figsize=(6,4))
plt.plot(month_counts.index, month_counts.values, marker='o')
plt.title("Month-wise Content Addition on Netflix")
plt.xlabel("Month")
plt.ylabel("Number of Titles Added")
plt.xticks(range(1,13))
plt.tight_layout()
plt.show()
print("Month with Highest Additions:", month_counts.idxmax())
```



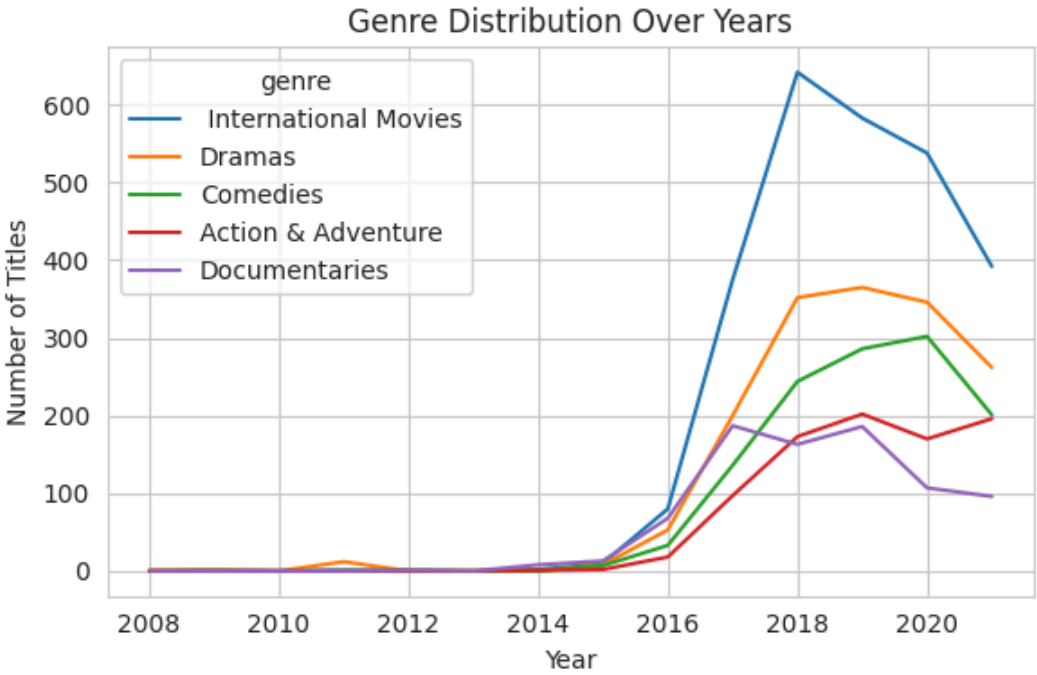

```
Month with Highest Additions: 7.0
```

Insight:

- Netflix adds content throughout the year, but certain months show higher activity.
- Peak months usually occur during mid-year and festive seasons.
- Increased content addition during these periods helps maximize viewer engagement.
- Strategic timing aligns with holidays and higher user activity.
- Regular monthly additions ensure continuous user interest and platform freshness.

Q19- How does the genre distribution vary across different years?

In [56]:
```
df['year_added'] = df['date_added'].dt.year
df['genre'] = df['listed_in'].str.split(',')
df_exploded = df.explode('genre')
genre_year = df_exploded.groupby(['year_added', 'genre']).size().unstack().fillna(0)
top_genres = df_exploded['genre'].value_counts().head(5).index
genre_year[top_genres].plot(figsize=(6,4))
plt.title("Genre Distribution Over Years")
plt.xlabel("Year")
plt.ylabel("Number of Titles")
plt.tight_layout()
plt.show()
```

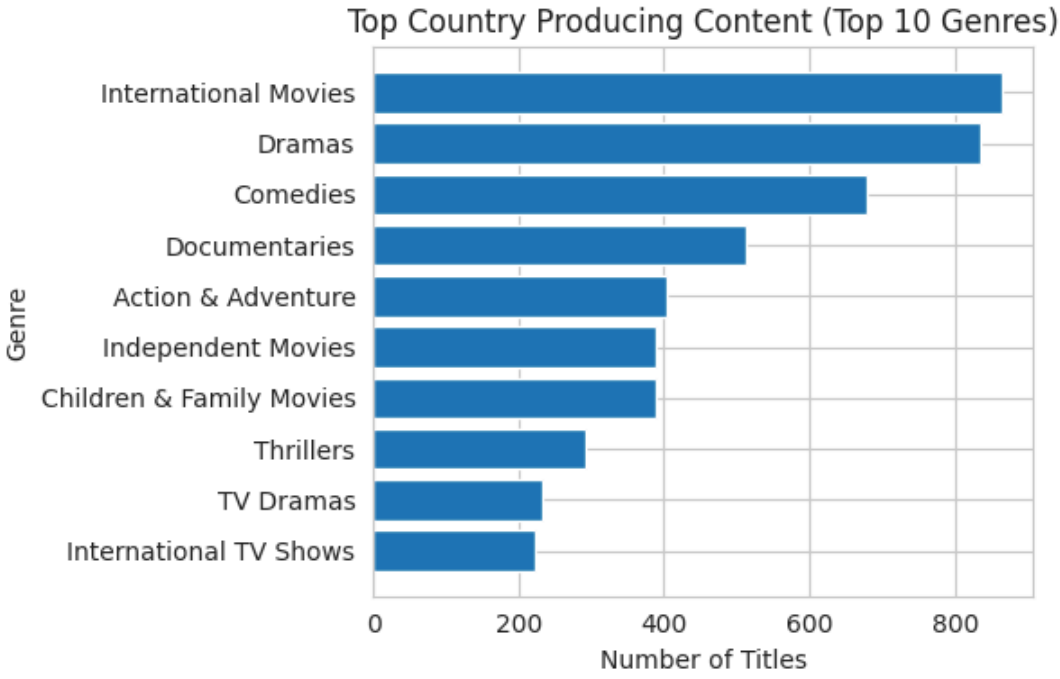


Insight:

- The diversity of genres has increased over the years, indicating content expansion.
- Traditional genres like Drama and Comedy remain consistently dominant.
- New genres such as Documentaries and Crime have gained popularity in recent years.
- The variation in genre distribution reflects changing audience preferences over time.
- Netflix continuously adapts its content strategy to include a wider range of genres for global audiences.

Q20- Which countries produce the most content in each genre?

In [57]:
```
df['country_split'] = df['country'].str.split(',')
df['genre_split'] = df['listed_in'].str.split(',')
df['country_split'] = df['country_split'].apply(lambda x: [i.strip() for i in x])
df['genre_split'] = df['genre_split'].apply(lambda x: [i.strip() for i in x])
df_exploded = df.explode('country_split').explode('genre_split')
top_genres = df_exploded['genre_split'].value_counts().head(10).index
filtered = df_exploded[df_exploded['genre_split'].isin(top_genres)]
genre_country = filtered.groupby(['genre_split', 'country_split']).size().reset_index(name=
top_country_genre = genre_country.sort_values(['genre_split','count'], ascending=[True, Fal
.groupby('genre_split').head(1)
top_country_genre = top_country_genre.sort_values('count')
plt.figure(figsize=(6,4))
plt.barh(top_country_genre['genre_split'], top_country_genre['count'])
plt.title("Top Country Producing Content (Top 10 Genres)")
plt.xlabel("Number of Titles")
plt.ylabel("Genre")
plt.tight_layout()
plt.show()
```



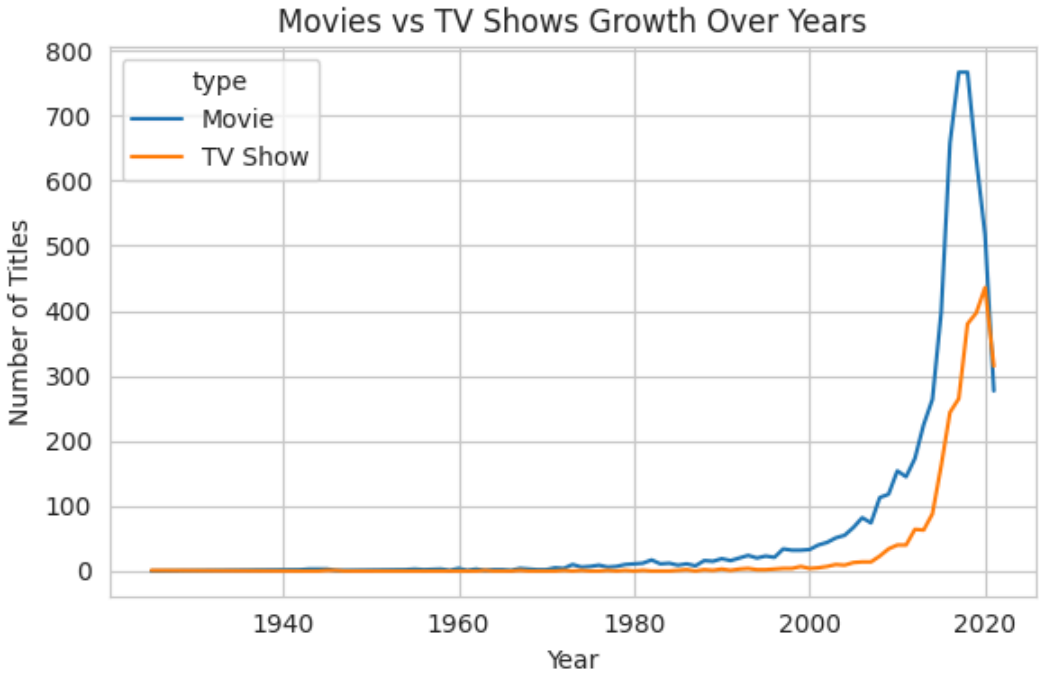
Insight:

- The United States dominates production across most genres, highlighting its strong entertainment industry.
- Different countries specialize in specific genres based on cultural and regional preferences.
- India contributes significantly to genres like Drama and Romance.
- International diversity in genre production helps Netflix cater to a global audience.
- Country-genre relationships indicate opportunities for localized content strategies and market expansion.
- Data was preprocessed by removing extra spaces to ensure accurate grouping and visualization.

Additional Analysis-

1. Content Growth Comparison (Movies vs TV Shows over Years}.

In [58]:
```
type_year = df.groupby(['release_year', 'type']).size().unstack().fillna(0)
type_year.plot(figsize=(6,4))
plt.title("Movies vs TV Shows Growth Over Years")
plt.xlabel("Year")
plt.ylabel("Number of Titles")
plt.tight_layout()
plt.show()
```

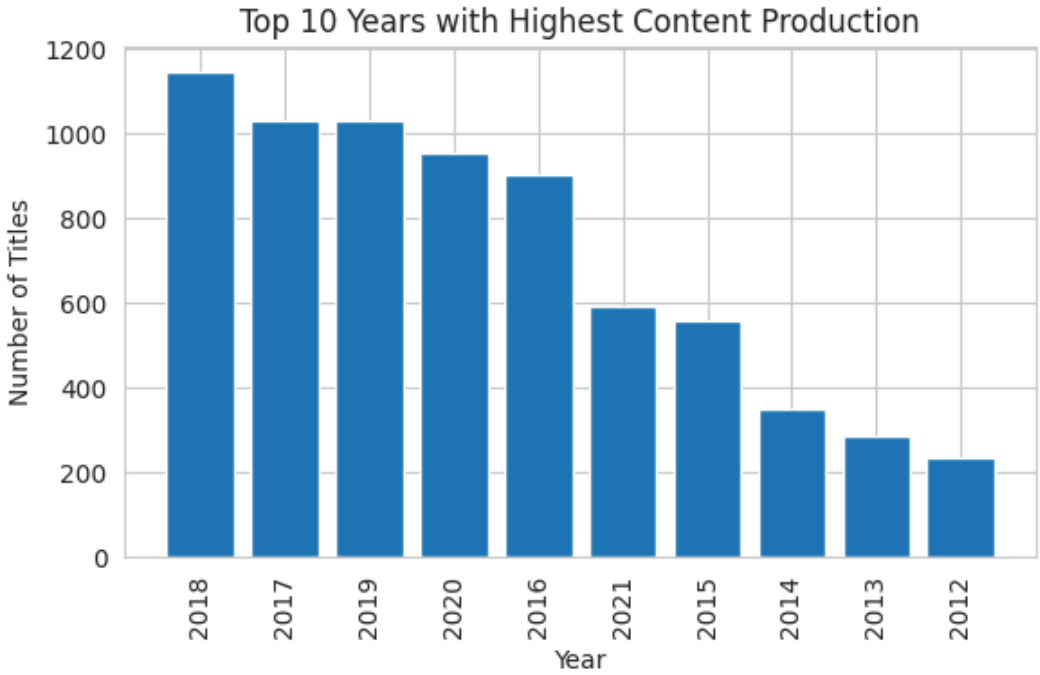


Insight:

- Both Movies and TV Shows have increased significantly over time.
- Movies show a consistently higher count compared to TV Shows.
- TV Shows have grown rapidly in recent years, indicating increased investment in series content.
- The trend reflects Netflix's strategy of balancing short-term (movies) and long-term (TV shows) engagement.
- Growth in both categories aligns with Netflix's expanding global user base.

2. Top 10 Content-Producing Years (Peak Years Analysis).

In [59]:
```
top_years = df['release_year'].value_counts().head(10)
plt.figure(figsize=(6,4))
plt.bar(top_years.index.astype(str), top_years.values)
plt.title("Top 10 Years with Highest Content Production")
plt.xlabel("Year")
plt.ylabel("Number of Titles")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```

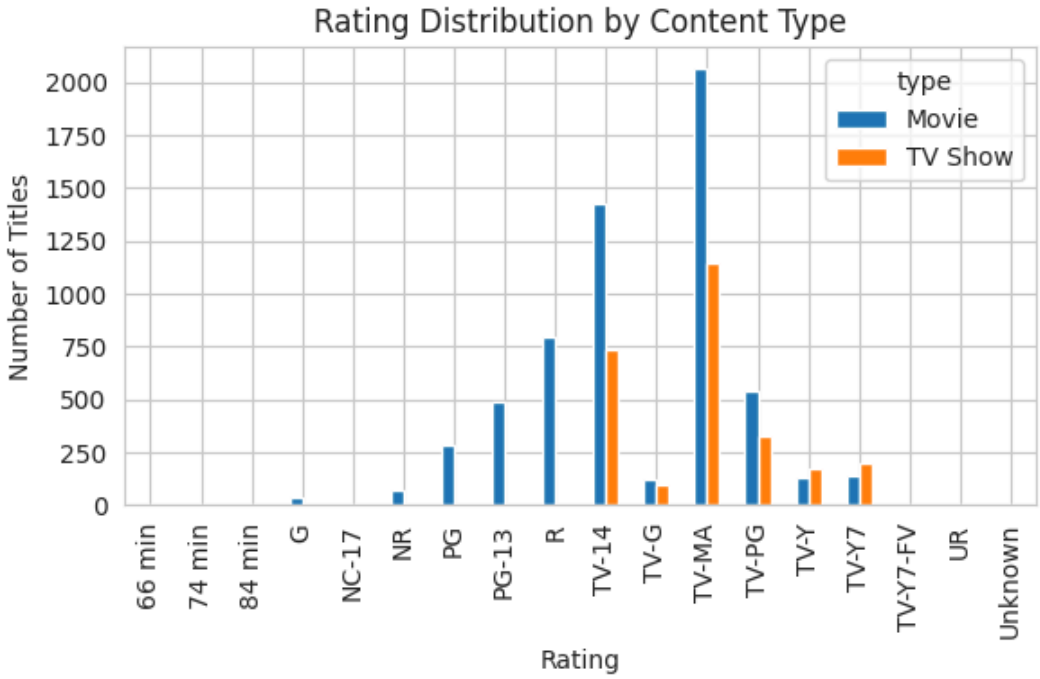


Insight:

- Certain years show peak content production, highlighting Netflix's aggressive expansion phases.
- Recent years dominate the top list, indicating rapid growth in content creation.
- These peak years align with increased competition in the streaming industry.
- Higher production reflects Netflix's strategy to capture and retain users.
- Content spikes suggest heavy investment in original programming.

3. Rating vs Content Type (Movies vs TV Shows Audience Targeting).

In [60]:
```
rating_type = df.groupby(['rating', 'type']).size().unstack().fillna(0)
rating_type.plot(kind='bar', figsize=(6,4))
plt.title("Rating Distribution by Content Type")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



Insight:

- Movies dominate across most rating categories compared to TV Shows.
- TV Shows have a higher concentration in specific ratings like TV-MA and TV-14.
- The distribution shows Netflix targets both mature and younger audiences.
- Movies are more diverse in audience targeting compared to TV Shows.
- This analysis highlights Netflix's strategy of content diversification across age groups.

Summary of Key Findings:

- Movies dominate Netflix content, but TV Shows play a crucial role in long-term user engagement.
- Drama, Comedy, and International genres are the most popular across the platform.
- Content production and addition increased significantly after 2015, reflecting Netflix's global expansion.
- The United States is the leading content producer, followed by countries like India and the UK.
- Mature content (TV-MA) is highly prevalent, indicating strong demand for adult-oriented storytelling.
- Most TV Shows have 1–2 seasons, showing a trend toward short-format content.
- Popular actors and directors frequently collaborate with Netflix, contributing to content success.
- Genre diversity has increased over time, reflecting evolving audience preferences.

Limitations of the Study:

- The dataset is limited to the available Netflix data and may not include the most recent content updates.

- Some fields such as director, cast, and country contain missing or “Unknown” values, which can affect analysis accuracy.
- The analysis is based only on descriptive statistics and visualizations; advanced techniques like machine learning were not applied.
- Multi-genre and multi-country entries were simplified during analysis, which may not fully capture complex relationships.
- The dataset does not include user viewing data, ratings, or watch time, limiting insights into actual audience behavior.
- External factors such as marketing, competition, and platform algorithms are not considered in this analysis.
- The results are based on historical data and may not fully represent future trends.

Final Conclusion:

- Netflix primarily focuses on movies, while TV Shows are used for long-term user engagement and retention.
- Drama, Comedy, and Thriller are the most dominant genres, reflecting strong audience demand.
- Content production and addition have increased significantly after 2015, indicating rapid global expansion.
- The United States is the leading content producer, followed by countries like India and the UK.
- A high proportion of TV-MA content shows growing demand for mature and realistic storytelling.
- Most TV Shows have 1–2 seasons, highlighting the popularity of short-format content.
- Netflix is increasingly investing in international and region-specific content to attract global audiences.
- Movie durations remain consistent, showing stable audience viewing preferences.
- Strong collaborations with directors and actors play a key role in content success.
- Overall, Netflix’s growth depends on content diversity, strategic planning, and data-driven decision-making.
- These insights can help Netflix make strategic, data-driven decisions to improve content production and global growth.
- Additional analysis reinforces the observed trends and provides deeper insights into Netflix’s content strategy and audience targeting.

Recommendations:

- Netflix should continue investing in popular genres like Drama, Thriller, and Comedy to maximize viewer engagement.
- Increase production of international and regional content to attract diverse global audiences.
- Focus on short-format TV Shows (1–2 seasons) to match modern viewing habits and reduce production risk.

- Strengthen partnerships with frequently appearing directors and actors to maintain content quality.
- Improve data quality by reducing missing values in fields like director, cast, and country.
- Use data-driven insights to plan content release timing, especially during high engagement periods.
- Expand into emerging markets by producing culturally relevant and localized content.

Future Scope:

- The analysis can be extended using more recent Netflix datasets for updated insights.
- Machine learning models can be applied to predict content success and viewer preferences.
- Advanced analysis can be done using user behavior data such as watch time and ratings.
- Sentiment analysis on content descriptions can help understand audience interest.
- A recommendation system can be developed to personalize user experience.
- Comparative analysis with other streaming platforms (like Amazon Prime, Disney+) can provide deeper insights.
- Further research can explore trends in specific genres, actors, or regions over time.